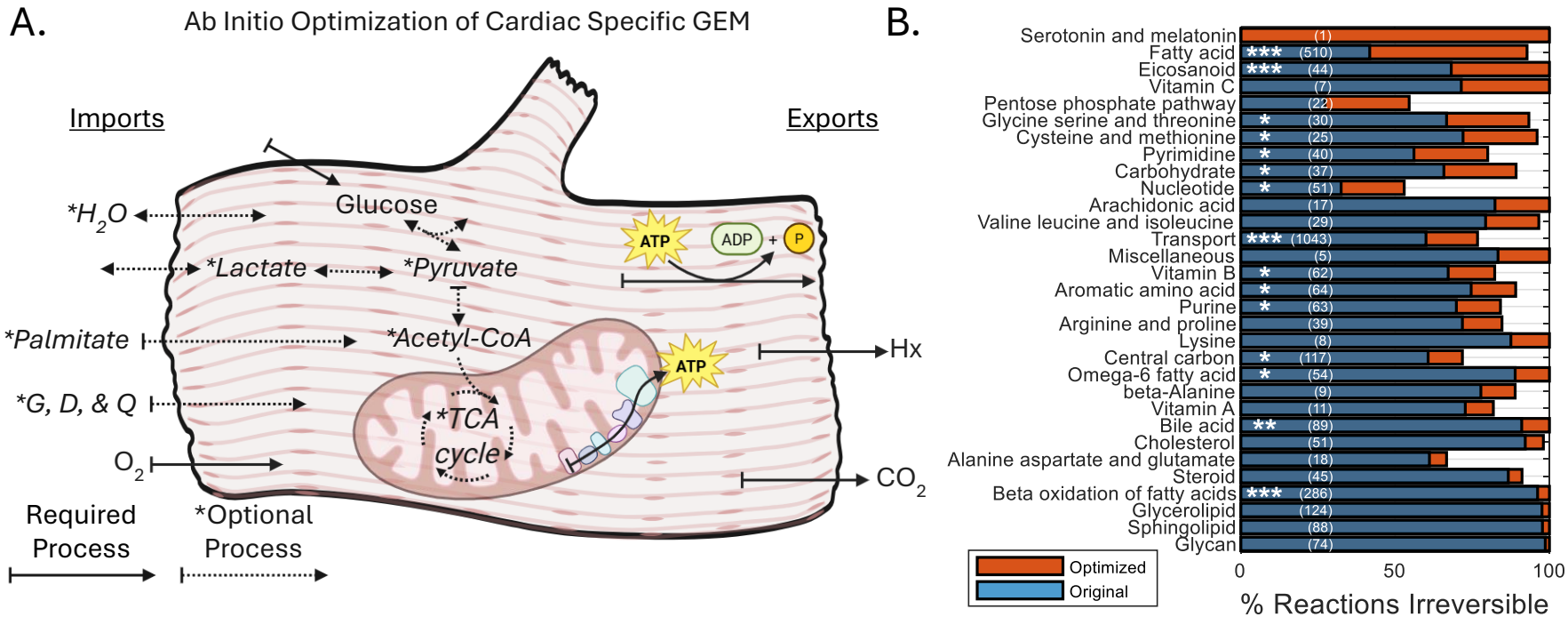
A.
Ab Initio Optimization of Cardiac Specific GEM
Imports
Exports
*H2O
Glucose
*Lactate
*Pyruvate
ATP
ADP
+
P
*Palmitate
*Acetyl-CoA
ATP
Hx
*G, D, & Q
*TCA
cycle
O2
CO2
Required
Process
*Optional
Process
B.
Serotonin and melatonin (1)
Fatty acid *** (510)
Eicosanoid *** (44)
Vitamin C (7)
Pentose phosphate pathway (22)
Glycine serine and threonine * (30)
Cysteine and methionine * (25)
Pyrimidine * (40)
Carbohydrate * (37)
Nucleotide * (51)
Arachidonic acid (17)
Valine leucine and isoleucine (29)
Transport *** (1043)
Miscellaneous (5)
Vitamin B * (62)
Aromatic amino acid * (64)
Purine * (63)
Arginine and proline (39)
Lysine (8)
Central carbon * (117)
Omega-6 fatty acid * (54)
beta-Alanine (9)
Vitamin A (11)
Bile acid ** (89)
Cholesterol (51)
Alanine aspartate and glutamate (18)
Steroid (45)
Beta oxidation of fatty acids *** (286)
Glycerolipid (124)
Sphingolipid (88)
Glycan (74)
Optimized
Original
0
50
100
% Reactions Irreversible